Feature glossary — voice pipeline

Plain-language description of each of the 24 features in `feature_summary.md` / `author_feature_results.csv`. Grouped by type; original computation is in `features.py` if you need the exact formula.

Vocabulary / word-choice features

Feature	What it measures
`ttr`	Type-token ratio: unique words ÷ total words in the window. Higher = more varied vocabulary, less repetition. (Note: mechanically shrinks as window size grows — see README caveat.)
`hapax_rate`	Share of words that appear exactly once in the window ("hapax legomena"). Higher = more one-off, unusual word choices rather than a small set of words repeated often.
`avg_word_len`	Average word length in characters. Higher = longer, often more Latinate/formal vocabulary.
`syll_per_word`	Average syllables per word. A rough proxy for vocabulary complexity/formality.
`pct_monosyllabic`	Share of words that are exactly one syllable. Higher = plainer, blunter word choice.

Specific function words

These track how often one particular common word appears, as a rate per 100 words. Function words (articles, prepositions, conjunctions, pronouns) are a classic stylometry signal because authors use them unconsciously and consistently, unlike topic words which change with subject matter.

Feature	Word tracked
`fw_the`	"the"
`fw_and`	"and"
`fw_of`	"of"
`fw_to`	"to"
`fw_in`	"in"
`fw_that`	"that"
`fw_it`	"it"
`fw_was`	"was"
`fw_but`	"but"
`fw_as`	"as"

Sentence structure

Feature	What it measures
`sent_len_mean`	Average sentence length, in words.
`sent_len_std`	How much sentence length varies within the window — a mix of short punchy and long winding sentences vs. a more uniform rhythm.

`subordination_rate`	Rate (per 100 words) of subordinating conjunctions — "because," "although," "since," "while," "if," "when," "though," "unless," "whereas," "whenever," "after," "before," "until," "once." Higher = more dependent-clause, "although X, Y" style sentences.
`coordination_rate`	Rate (per 100 words) of coordinating conjunctions — "and," "but," "or," "nor," "for," "yet," "so." Higher = more clauses strung together as equals ("X, and Y, and Z") rather than subordinated.

Punctuation

Feature	What it measures
`comma_rate`	Commas per 100 words.
`semicolon_rate`	Semicolons per 100 words.
`dash_rate`	Dashes per 100 words. (Flagged as unstable in the corpus-wide run — rare enough that some authors' reference books have very few, inflating their z-scores. Treat results on this one cautiously.)
`question_rate`	Share of sentences ending in "?".
`exclaim_rate`	Share of sentences ending in "!".

Reading the summary table

For each feature, `feature_summary.md` reports, out of the 33 authors run: how many showed that feature as a significant **burst** (spikes in the opening pages, then settles down), how many as a significant

accumulator (builds gradually across the whole book, no early spike), how many as **both**, and how many as **null** (nothing significant either way) — plus which of burst/accumulator/both was most common among the authors that showed *any* significant pattern (`dominant_signal`).